16

TREE OF LIFE



Observe the poster displayed at the science fair. What is the main idea conveyed by the poster?

Can you suggest a suitable title for the poster?

How is it possible to identify and study so many living organisms?

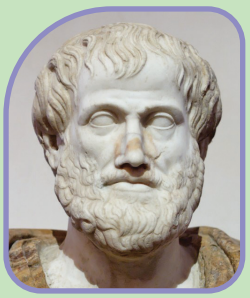
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Attempts to classify and study living organisms began a long time ago.

Let us get to know some personalities who made significant contributions in this field.

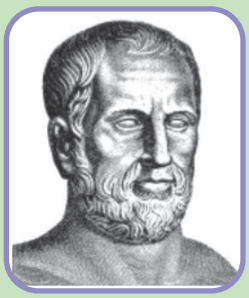
In the early days, organisms were classified as useful and non useful. Later, scientists tried to classify living organisms in various ways.

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Aristotle classified living beings as those with red blood and those without.

Figure 16.1

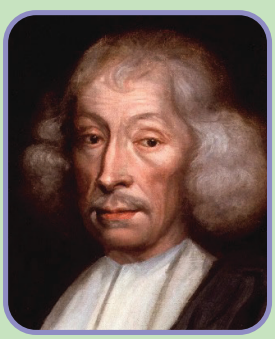


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Theophrastus classified plants into trees, shrubs, and herbs.

Figure 16.2

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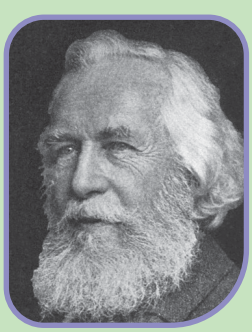
John Ray, who lived in the 17th century, initiated the scientific method of classification and was the first to use the term "species."

Figure 16.3

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Ernst Haeckel, who lived in the 19th century, classified living organisms into three kingdoms : Animalia, Plantae and Protista.

Figure 16.4



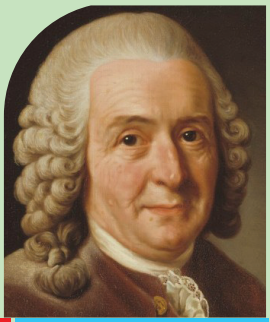


Figure 16.5

Carl Linnaeus, who lived in the 18th century, was the one who introduced the different levels of classification. Since he refined these classification levels and provided a scientific foundation for classification, he is considered the Father of Taxonomy.

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Many scientists have contributed to this field. Collect more information and prepare a wall poster titled "History of Taxonomy".

The classification of living beings was based on the similarities and differences in their characteristics.

Taxonomy is the branch of science that mainly focuses on classifying organisms by comparing their characteristics, identifying similarities and differences and giving them scientific names. Through taxonomy, we can understand biodiversity and the relationships between different organisms.

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In the classification of Linnaeus, the level that includes all multicellular heterotrophic organisms is Kingdom Animalia. The level just below the kingdom is called Phylum. All vertebrates in the kingdom Animalia are included in the Phylum Chordata. The most basic level of classification is Species.

A species is a group of organisms that can naturally reproduce fertile offspring. Human beings, cats, dogs, and cashew trees are examples of different species. What are the classification levels between species and phylum? Refer the illustration 16.1.

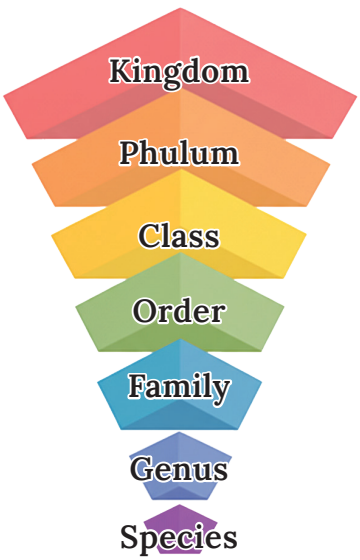


Illustration 16.1

Levels of classification

Identification, classification and naming of organisms are the main steps in classification.

Observe the picture given and complete Table 16.1 by verifying the classification levels of the tiger.



Figure 16.5



Figure 16.6 The classification levels of the tiger.

Levels of classification	Group and characteristics	Organisms
Kingdom:	Animalia (Animals) – Multicellular, eukaryotic, heterotrophic organisms.	
Phylum:	Chordata (Vertebrates) – Presence of vertibrae and vertebral column.	
Class:	Mammalia (Mammals) – Warm-blooded, body covered with hair, nourish young ones with milk.	
Order:	Carnivora (Carnivores) – Sharp teeth and claws adapted for hunting and eating meat.	
Family:	Felidae – Retractable claws, excellent night vision.	
Genus:	Panthera (Big cats) – Ability to roar, large and powerful body structure.	
Species:	Panthera tigris (Tiger) – Orange fur with black stripes and strong large body	

Table 16.1

The classification levels developed by Carl Linnaeus are still in use today. In classification, we group organisms into large groups based on similarities and into smaller groups based on differences. These classification levels such as kingdom, Phylum, Class etc. are called Taxa. Just like animals plants can also be classified.

It was Carl Linnaeus who classified plants by including the levels from species to kingdom.

Observe Table 16.2 which shows the classification levels of the coconut tree.

Kingdom:	Plantae (Plants) – Multicellular, eukaryotic, autotrophic organisms.
Division:	Magnoliophyta (Flowering plants) – Plants with seeds enclosed inside fruits.
Class:	Liliopsida (Monocotyledons) – Plants with parallel leaf venation and fibrous roots.
Order:	Arecales – Plants with long and hard stems.
Family:	Arecaceae – Plants with large compound leaves and unbranched stems.
Genus:	Cocos – Produces large, single-seeded fruits.
Species:	Cocos nucifera (Coconut) – Produces fruits with a hard shell, fibrous husk, with a liquid and solid endosperm, inside.

Table 16.2



Classification levels of coconut tree

Figure 16.7

Observe and identify one or two plants or animals found in your surroundings and find out their classification levels. Present your findings in class and compile the collected information into a well-organized classification register.

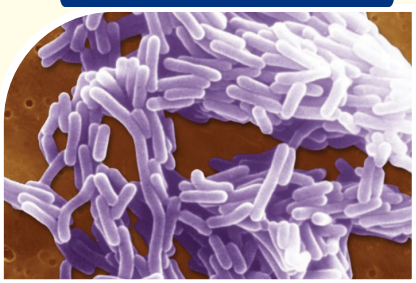
According to Carl Linnaeus' classification system, living organisms were divided into two kingdoms: Animalia and Plantae. This was known as the two-kingdom classification system.

Neither a plant nor an animal!

Can we include the organisms shown in Figure 16.8 (a, b, c, d) in the kingdoms we have studied earlier? Why?

Analyse the pictures and clues and write a suitable explanation for this.

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Bacteria

Figure 16.8 (a)

- Unicellular organism
- Prokaryote

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Paramecium

Figure 16.8 (b)

- Unicellular organism
- Eukaryote
- Heterotroph
- Have mobility

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Mushroom

Figure 16.8 (c)

- Eukaryote
- Have no mobility
- Heterotrophs



Yeast

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Figure 16.8 (d)

Five Kingdom classification

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Bacteria, Paramecium, Mushroom and Yeast cannot be included in the Plant or Animal kingdoms. To overcome this limitation of the two-kingdom classification, the scientist Robert H. Whittaker classified organisms into five kingdoms.

Monera, Protista, Fungi, Plantae and Animalia are the five kingdoms. Analyse the description of the five-kingdom classification and complete Table 16.3 appropriately.



Robert H. Whittaker

Figure 16.9

Kingdom Monera	Prokaryotes, do not have a distinct nucleus. They include some of the most ancient living organisms. Example: Bacteria.
Kingdom Protista	Most of them are unicellular eukaryotes. They can be autotrophs or heterotrophs. Flagella, cilia, and pseudopodia are their locomotory structures. Eg. Amoeba, Paramecium, and Euglena are examples.
Kingdom Fungi	Eukaryotes that are saprophytes. They do not have chlorophyll. Examples are mushrooms and yeast.
Kingdom Plantae	Multicellular organisms that produce their own food through photosynthesis, Eukaryotes. Primary producers. They have roots, stems and leaves. They grow in soil or water.
Kingdom Animalia	Multicellular eukaryotes. Heterotrophs. They have locomotory organs and definite receptors. They range from simple organisms like sponges to complex organisms like humans.

Kingdom	Characteristics	Examples
Animalia		
Plantae		
Monera		
Protista		
Fungi		

Table 16.3

Find out the advantage of the five-kingdom classification.

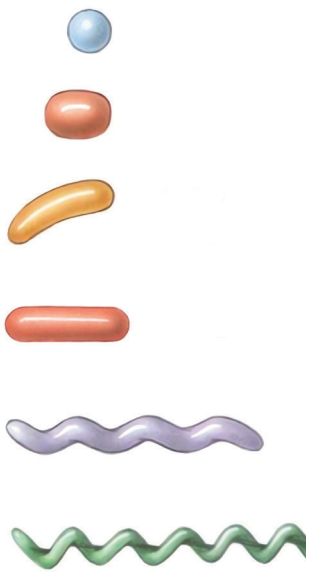
Monera, Protista, and Fungi are groups that include simple microorganisms. However, Plantae and Animalia include organisms with more complex structures.

Monera

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3D VIEW

Monera is the group of the simplest and most ancient organisms. They are all unicellular and prokaryotic. The main organisms included in this group are bacteria (Figure 16.10). Bacteria can be found in soil, water, air, food, and even inside our bodies. They play an important role in nutrient cycles like nitrogen fixation in the soil. Although most bacteria are useful, some can cause diseases.



Bacteria of various shapes

Figure 16.10

Human Microbiome Project (HMP)

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The Human Microbiome Project is a research initiative launched in 2007 by the U.S. National Institutes of Health (NIH) to study microbial communities related to human health and disease. The human microbiome includes microorganisms such as bacteria, viruses, and fungi which live in and outside the human body. These play a vital role in maintaining health and influence digestion, immunity, and even mental well-being.

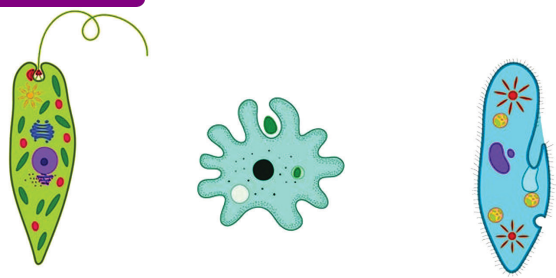
Protista

CLICK TO SEE THE VIDEO

3D VIEW

These are unicellular eukaryotes. Most protists live in aquatic habitats. Paramecium, Amoeba, and Euglena are examples.

Plasmodium and Entamoeba histolytica are disease-causing protists.



Figures 16.11, 16.12, 16.13

Fungi

CLICK TO SEE THE VIDEO

3D VIEW

These are eukaryotes. Except yeast, the others are multicellular.



Figure 16.14

Fungal cells have a cell wall made of a substance called chitin. Since fungi do not have chlorophyll, they cannot perform photosynthesis. They absorb nutrients from dead and decaying matter, so they are called saprophytes.

Mould on bread, mushrooms, and yeast are examples of fungi. Edible mushrooms, yeast used in baking and preparation of wine and alcohol are useful fungi. Some fungi cause diseases such as candidiasis and ringworm.

Behind the name



Figure 16.15

What is this plant known as in your area?

Papaya, karumoosa, kappalanga, ...

You've understood now that in different places, the same plant has different names. Even in Malayalam, there are various names, so it's certain that in different languages, it will have different names too.Wouldn't this make it difficult to identify and study the same living organism when it is known by different names?

What could be the solution to this?

Write down your guess.

.....

This issue is solved by giving names that are universally accepted beyond language barriers.

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It was Carl Linnaeus who developed the method of naming organisms scientifically. This scientific method of naming where the genus name comes first and the species name second is called Binomial nomenclature.

When a name is given in this manner, the organism will have the same name everywhere in the world. Accordingly, the scientific name of human is Homo sapiens.

The names of living organisms are mentioned in Latin. The names in other languages are transformed to Latin style to give scientific names to organisms.

Rules of Binomial Nomenclature

In the scientific naming system, the name of an organism should have two parts. The first part of the name is the genus name and the second part is the species name. When writing the scientific name in English, the genus name should begin with a capital letter. The scientific name should be printed in italics when typed.

Find the scientific names of the organisms mentioned below and complete Table 16.4.

Organism	Scientific name
Tiger	
Coconut tree	
Dog	

Table 16.4

New paths in classification

As new organisms are discovered, the existing classification systems become insufficient to accurately categorize them. Classification methods are continuously updated based on physical characteristics, genetic relationships, and other factors. Classification is constantly being revised in accordance with new observations, discoveries and understandings.

The methodology of modern classification

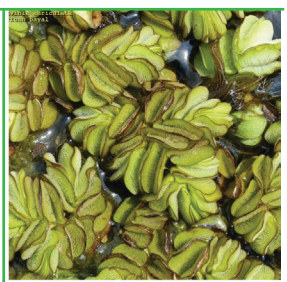
In the early days, organisms were classified based on their external structure and utility. What is the limitation of such a method of classification?



Azolla



Wolffia



Salvinia

Figure 16.16

Can the simply structured Salvinia and Wolffia be classified under the same genus?

Based on the indicators given, examine the description below.

According to modern classification methods, they do not even belong to the same family. The non-flowering Salvinia belongs to the family Salviniaceae which includes plants like Azolla. On the other hand, Wolffia is classified under the family Araceae which includes flowering plants like Anthurium. Both Wolffia and Anthurium are flowering plants and are believed to have evolved from a common ancestor. This evolutionary relationship is the reason they are placed in the same family, despite their large external differences.

Similarly, classifications based only on external features may lead to genetically related organisms being placed in different groups. Today, organisms are classified by considering both evolutionary (phylogenetic) and genetic characteristics.

Indicators

- Limitations of classification based only on external features
- Changes in the criteria of classification

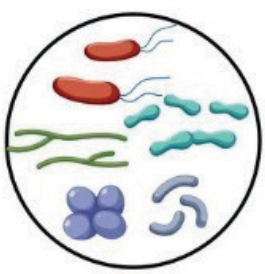
Six-Kingdom classification



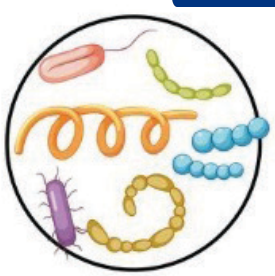
Figure 16.17

The scientist Carl Woese introduced the six-kingdom classification system. The kingdom Monera was divided into two distinct kingdoms to form this system. Based on modern scientific observations, the kingdom Monera was split into two groups: Archaebacteria and Eubacteria.

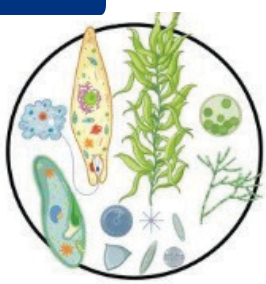
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Archaebacteria



Eubacteria



Protista



Fungi



Plantae



Animalia

Illustration 16.2

Do all humans belong to the same species?

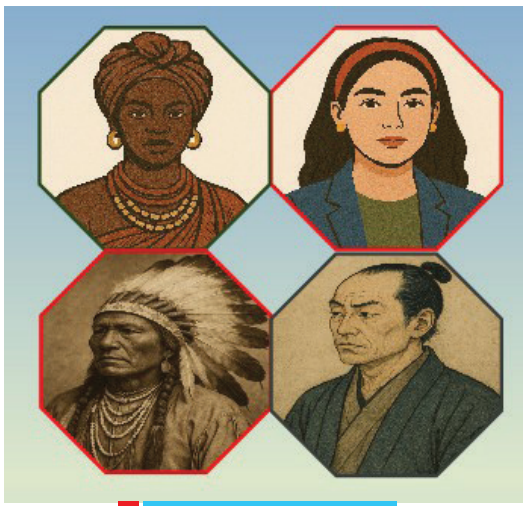


Illustration 16.3

Although humans living in different parts of the earth show some differences in their external appearance due to the influence of local geography and environmental factors, there is no significant genetic or other fundamental difference among them. All humans around the world share a common ancestor. That is why all humans are classified under the same species, sapiens.

As science progresses, advancements in molecular-level studies have led to changes and improvements in classification systems. With the discovery of each new organism, taxonomy continues to evolve. This process deepens our connection with nature.

Let's Assess

1. Examine the following passage and correct the errors if any, found in the underlined parts.

   Classifying living beings into larger groups based on similarities and further into smaller groups based on differences is the basis of the cell theory. The various levels of classification were developed by Theophrastus. He is known as the Father of Taxonomy. The smallest

unit in classification is phylum. The method of scientifically naming organisms was developed by Carl Linnaeus. The practice of identifying an organism by using both its genus and species name is known as taxon. According to this, when writing the scientific name of an organism, the species name should be written first.

2. Match the following.

A	B
1. John Ray	a) Living things were classified into 5 kingdoms
2. Carl Linnaeus	b) Plants were classified as trees, shrubs, and herbs
3. Ernst Haeckel	c) Living things were classified into 3 kingdoms
4. Robert H. Whittaker	d) Father of taxonomy
5. Aristotle	e) Initiated the scientific method of classification
	f) Animals were classified into those with red blood and those without red blood.

Example: Aristotle classified animals into two groups – those with red blood and those without red blood.

3. Which of the following does not belong to the group? What is the common feature of the others?

A. Housefly, Lizard, Butterfly, Grasshopper

B. Cat, Dog, Tiger, Snake

C. Arecanut tree, Coconut tree, Jack tree, Palm tree

Extended Activities

1. Prepare a magazine including pictures and contributions of scientists related to the history of classification.
2. List the plants found in the school surroundings and biodiversity parks, find their scientific names, and display them.